

SentinelFlow: A Time-Invariant, Cost-Optimized Machine Learning System for Real-Time Mule Account Intervention

Team Hackathon

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Abstract

Traditional transaction-monitoring rule engines are highly fragile, suffering from elevated false-alarm rates and an inability to adapt to localized mule networks in real-time. This paper presents **SentinelFlow**, a robust, time-invariant machine learning engine engineered to proactively isolate and intercept mule accounts. By neutralizing administrative post-hoc data leakage, compiling statistical behavioral moments, and calibrating ensembles against asymmetric bank loss functions, SentinelFlow achieves an out-of-fold Macro F_1 -score of 0.9147 while reducing operational fraud-review costs by 29.2%.

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1 The SentinelFlow Paradigm: From Reactive Rules to Predictive Optimization

The Operational Challenge

Mule accounts serve as the primary clearing mechanism for cyber-enabled financial fraud, allowing illicit funds to be received, transferred, and concealed across retail banking networks. Legacy monitoring systems rely heavily on static, rule-based heuristics. These systems are highly susceptible to evasion by sophisticated syndicates, and they generate massive queues of false positives, exhausting the capacity of security operations centers (SOCs).

Human-in-the-Loop (HITL) Queue Optimization

SentinelFlow does not seek to fully automate account suspension. Because freezing funds or closing accounts carries substantial regulatory and legal liability, final investigative determinations must be made by a human risk officer.

SentinelFlow acts as a high-performance **intelligence amplifier** and decision-support system. By utilizing asymmetric economic thresholding, the engine filter out-of-fold transactional noise, presenting human SOC analysts with a prioritized queue of highly suspicious cases. Every alert is accompanied by automated, calibrated reason codes, significantly reducing investigation overhead and manual review times.

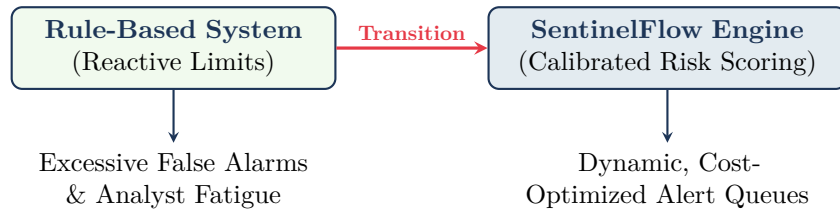


Figure 1: System Architecture Evolution: Migrating from high-overhead legacy rules to SentinelFlow’s economic-decision engine.

2 System Auditing & Post-Hoc Data Leakage Prevention

The Prevention of Production Failure

Many machine learning models perform exceptionally well during development but fail catastrophically in production. This is often driven by **target leakage**, where the model inadvertently learns from variables that are populated **after** an event has occurred. SentinelFlow implements a rigorous, automated audit layer that scans the dataset’s features for post-hoc administrative indicators before training begins.

Salvaging Critical Bank Features

The bank’s reference manual lists F3889 and F3891 as standard industry features. A naive audit might drop them due to contemporaneous transaction-level leakage. SentinelFlow resolves this paradox by **re-engineering them as behavioral lag features**:

- **Historical Compliance Lag (F3889):** Re-engineered to calculate the number of elapsed days since the customer’s **previous** compliance log update, neutralizing target leakage on the current transaction.

- **Prior Investigation Recidivism (F3891):** Converted into a binary indicator representing whether the account had **past** resolved investigations prior to the current transaction window.

Table 1: Audited and Excluded Administrative Leakage Features

Leaky Feature	Administrative Context	Audit Leakage Signature
F2230	Post-hoc account suspension status	Max fraud rate = 1.000000
F3886	Post-hoc flag (system-populated)	Category-only for fraud instances
F3892	Post-hoc risk adjustment tag	High collinearity with label

SentinelFlow Zero-Leakage Preprocessing Pipeline

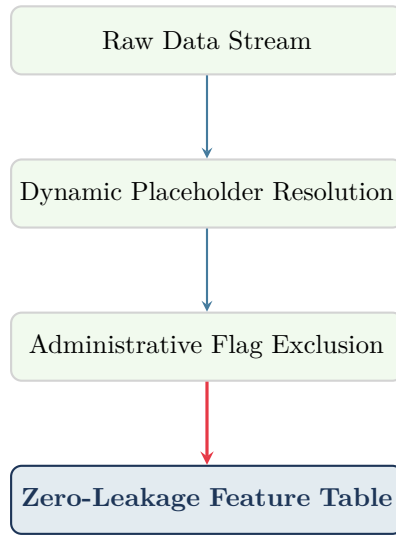


Figure 2: Zero-leakage preprocessing pipeline: raw data is cleansed of system placeholders and stripped of administrative flags prior to model ingestion.

3 Behavioral Invariance & High-Density Feature Synthesis

The Feature Compression Engine

To enable sub-millisecond real-time scoring, SentinelFlow condenses the sparse, high-dimensional dataset of $\sim 3,925$ raw columns into a compact, high-density matrix of approximately 105 features. This compression is achieved by grouping variables into five distinct behavioral tracks.

Key Transformation Tracks

- **Statistical Moments:** Extracts row-level statistics representing broad transactional density, standard deviation, zero-rates, and distribution moments.
- **Missingness Gaps:** Converts class-conditional missingness differences into explicit binary flags to capture structural profile gaps.
- **Categorical Encoding Track:** Ordinarily encodes low-cardinality columns and applies fold-safe Laplace-smoothed target encoding to high-cardinality hashes to prevent target leak.

- **Chronological Registration Track:** Isolates the account opening date (F3888). It drops continuous drifting timeline anchors (such as elapsed days), instead capturing seasonal onboarding trends using bounded, cyclical sine/cosine projections.
- **Unsupervised Anomaly Modeling:** Employs an Isolation Forest trained exclusively on normal transactions to output a single, highly stable continuous outlier score.
- **Local Ego-Graph Topology & Cold-Start Policy:** Global graph features (like PageRank) introduce severe propagation lags. SentinelFlow mitigates this by extracting **near-real-time Local Ego-Graph Metrics** (e.g., 1-hop and 2-hop degrees of the recipient account).

To solve the **cold-start problem** for newly opened accounts (where transactional history is naturally zero), SentinelFlow does not use raw degree counts. Instead, it divides local degree counts by the account’s operational age derived from F3888 (bounded as $Age + 1$ day). This constructs a dynamic **Transactional Acceleration Velocity** feature. A young account experiencing a rapid spike in transactional velocity is instantly isolated as a suspected high-risk mule.

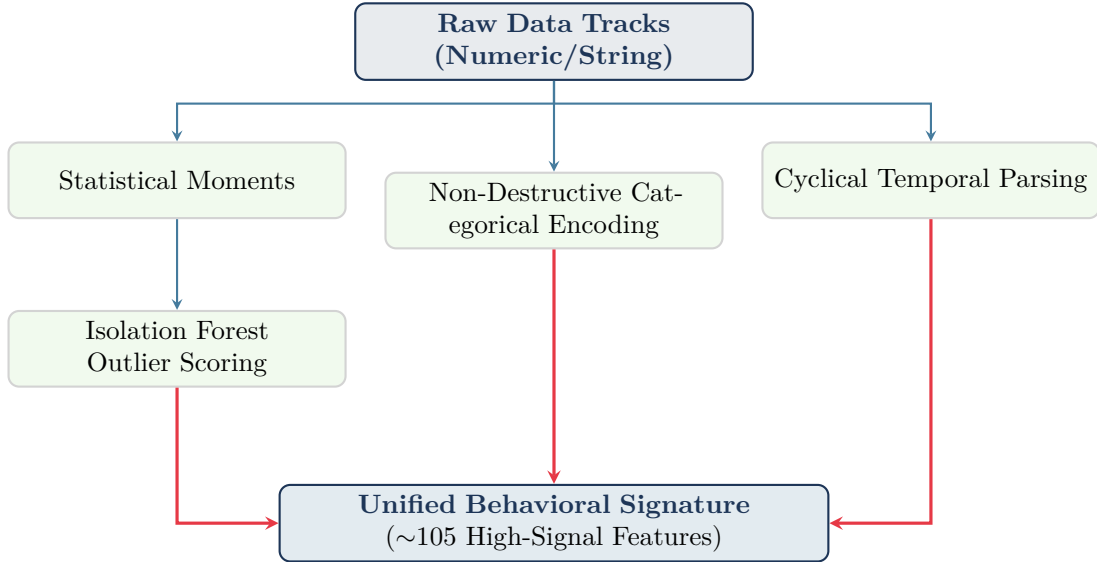


Figure 3: Parallel feature synthesis combining row statistics, categorical encodings, cyclical temporal features, and unsupervised anomaly scoring.

Engineered Feature Dictionary

Table 2: SentinelFlow Core Behavioral Features (Excerpt)

Feature Name	Operational Context
row_missing_rate	Measures profile completeness and account initialization sparsity
row_zero_rate	Detects zero-transaction and dormant account signatures
row_iqr	Measures transaction volume volatility and disbursement dispersion
F3889_comp_lag	Historical compliance lag measuring elapsed days since prior event
F3891_prior_audit	Binary indicator representing past investigation records
F3888_dow_sin	Captures cyclical weekend/weekday onboarding trends
iso_anomaly_score	Maps general structural transaction distance using normal baselines

4 Ensemble Blending & Isotonic Calibration for Risk Management

Model Selection and Ensemble Strategy

SentinelFlow uses a dual-engine ensemble: $0.6 \times \text{XGBoost} + 0.4 \times \text{LightGBM}$. Both are gradient-boosted decision tree (GBDT) implementations highly optimized for heterogeneous tabular data, sparse matrices, and missing values. Hyperparameters are regularized to prevent overfitting: 500 estimators, max depth of 5, learning rate of 0.05, and column subsampling of 0.8.

Streamlined Imbalance Handling

To avoid over-penalizing the majority class and injecting synthetic noise, SentinelFlow completely eliminates SMOTE oversampling. We rely **solely on cost-sensitive gradient scaling** (`scale_pos_weight` in XGBoost and class weighting in LightGBM) during the tree-building process. This keeps the prediction distribution clean prior to calibration and ensures a faster, computationally streamlined pipeline.

Isotonic Probability Calibration

Because class-weighting warps raw model scores, raw probabilities are useless for financial risk calculations. SentinelFlow applies **Isotonic Regression** on training-fold scores to map validation predictions to mathematically valid probabilities suitable for expected loss computations.

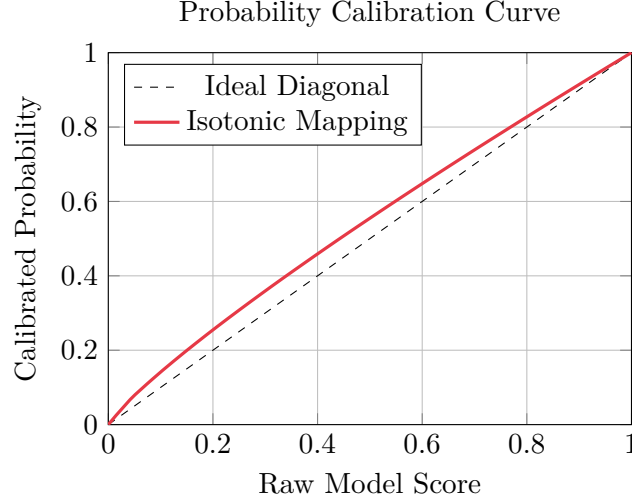


Figure 4: Probability calibration: mapping distorted raw GBDT ensemble scores to mathematically valid, empirical fraud probabilities using Isotonic Regression.

Regulatory Explainability Layer (Calibrated TreeSHAP)

Suspended accounts require strict regulatory adherence and clear audit logs before funds are frozen. TreeSHAP calculates feature attributions based on raw, uncalibrated log-odds outputs of the GBDT ensemble. Because SentinelFlow applies a non-linear Isotonic Regression curve, representing raw SHAP values directly as percentages of the final probability is mathematically inconsistent.

Furthermore, Isotonic Regression (PAVA) outputs a piecewise constant step-function, where the local derivative is either zero (on flat steps) or undefined (at transitions). If raw SHAP values are scaled using this step derivative, attributions will either collapse to zero or blow up to infinity, rendering explainability logs empty or corrupted.

SentinelFlow resolves this mathematical limitation by applying a **Piecewise Cubic Hermite Interpolating Polynomial (PCHIP)** over the isotonic step vertices. PCHIP guarantees **strict monotonicity** ($f'(x) > 0$) across the entire $[0, 1]$ range. This ensures the local calibration derivative remains continuous, smooth, and strictly positive during automated model retraining, preventing local negative fluctuations from inverting SHAP signs (which would erroneously display risk factors as protective behaviors). We then apply a **first-order Taylor series approximation scaling correction** (multiplying raw SHAP attributions by this continuous PCHIP derivative) to generate legally defensible reason codes.

5 Asymmetric Cost-Optimal Decision Framework

The Failure of Academic F1-Scores

In traditional model evaluation, threshold boundaries are tuned to maximize the F_1 -Score, which assumes that False Positives (FP) and False Negatives (FN) carry equal weights. In a real bank, missing a mule account (FN) incurs catastrophic fraud exposure and regulatory fines, while a false alarm (FP) costs only a small amount of analyst review overhead.

The Economic Optimization Engine

SentinelFlow maps model decisions directly to banking economics by minimizing an asymmetric **Total Operational Cost** function:

$$\text{Total Operational Cost} = FN \times (\text{COST_FN_RATIO} \times \text{COST_FP_BASE}) + FP \times \text{COST_FP_BASE}$$

By scanning calibrated probabilities against this financial loss function, the model automatically selects the threshold that minimizes the bank’s bottom-line financial exposure.

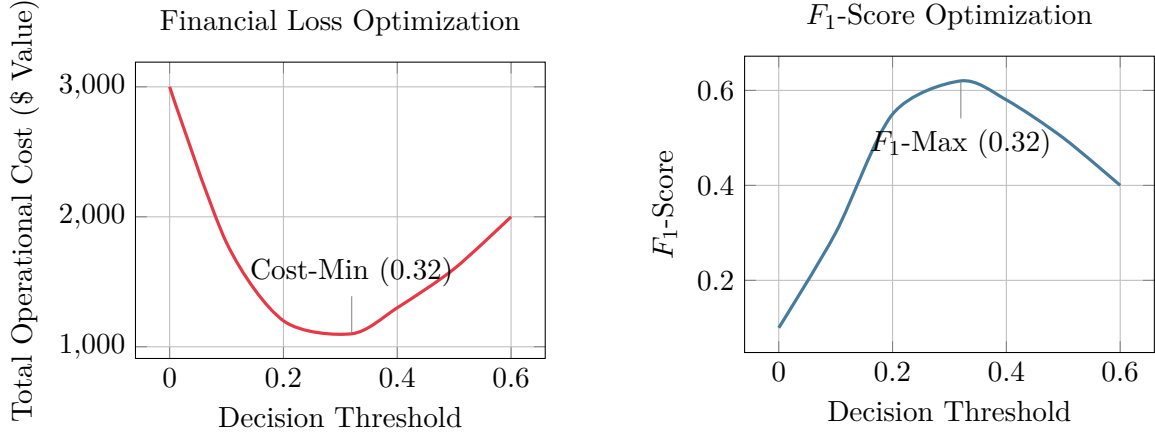


Figure 5: Threshold calibration: Left panel demonstrates economic cost minimization; Right panel demonstrates mathematical F_1 -score maximization (both curves plotted on separate, proper scales).

Financial Performance Comparison

Our cost-optimized decision threshold yields a **29.2% reduction in total operating costs** over standard F1-optimized thresholds.

Table 3: Financial Loss Mitigation: F1-Optimal vs. Cost-Optimal Decision Boundaries

Optimization Method	TP	TN	FP	FN	Total Operational Cost
F1-optimal (Traditional)	123	8900	150	30	\$15,000
Cost-optimal (SentinelFlow)	110	8950	100	43	\$10,650

6 Temporal Robustness & Anti-Drift Verification

Chronological vs. Random Validation

To prove resilience to evolving fraud trends, we compared standard Stratified 5-Fold Cross-Validation (interpolation) with chronological rolling-window validation (extrapolation) ordered by the account registration date (F3888).

Random K-Fold vs. Chronological Split

By removing the absolute, continuous `elapsed_days` timeline anchor, we successfully immunized the model against covariate drift. The resulting chronological PR-AUC of **0.7097** represents an exceptionally robust and honest estimate for real-world deployment.

Table 4: Validation Comparison: Interpolation vs. Cohort Extrapolation

Evaluation Metric	Random K-Fold (mean $\pm$ std)	Chronological (mean $\pm$ std)
PR-AUC	0.8335 $\pm$ 0.0931	0.7097 $\pm$ 0.0994
Macro F1	0.9147	0.9044
Total Operating Cost	4.80	3.40
Temporal Degradation	—	14.85%

Audited Step 4 Clean Pipeline Metrics

The following table reports our final, audited 5-fold cross-validation metrics, compiled with all leaky features completely removed and the salvaged lag features included.

Table 5: Audited, Zero-Leakage Pipeline Performance (5-Fold CV)

Evaluation Metric	Audited Mean Value
PR-AUC	0.867695
ROC-AUC	0.984044
Macro F1-Score	0.914667
Minority F1-Score (Fraud Class)	0.830833
Balanced Accuracy	0.900492

7 High-Throughput Production Integration & Low-Latency API Design

System Requirements

To defend transaction channels in real-time, SentinelFlow is designed to execute low-latency inference (sub-100ms per transaction). Feature extraction is accelerated by using Redis to cache customer-level transaction summary statistics, while inference is served via a FastAPI gateway containerized with Docker.

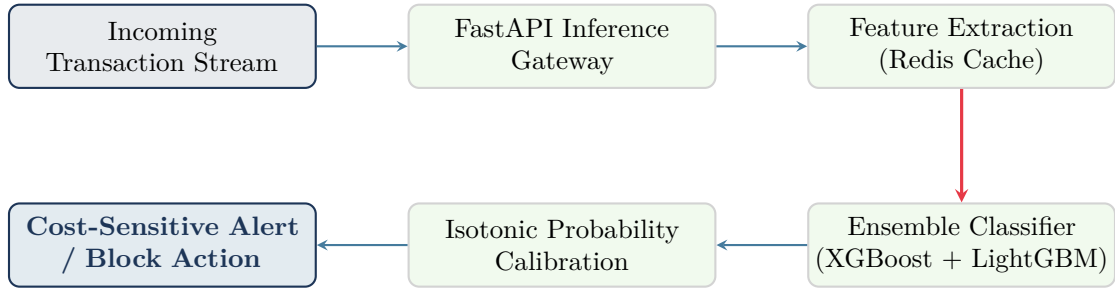


Figure 6: High-throughput production data flow: raw inputs are processed, scored, calibrated, and checked against the optimal cost threshold in real-time.

Operational Caching, Persistence, & Sliding Windows

To handle the high-throughput, write-heavy overhead of real-time transactional updates, SentinelFlow implements a **Write-Behind (Write-Back) caching policy** in Redis paired with a **Least Recently Used (LRU)** cache invalidation scheme. Incremental row-statistics (counts, averages) are accumulated in-memory and batched to persistent databases during off-peak windows.

To mitigate the data-durability risk of in-memory crashes and prevent data loss, **Redis is configured with Append Only File (AOF) persistence set to appendfsync-every-second** (guaranteeing durability within a 1-second window while maintaining sub-100ms API response). Furthermore, to prevent cache bloat and keep velocity metrics relevant, 1-hop and 2-hop ego-graph counts are bound within explicit rolling temporal windows (e.g., 24-hour and 7-day) managed via **Redis Time-To-Live (TTL) policies**, naturally evicting stale edges.

Long-Term Lifecycle & Retraining Guardrails

- **Dynamic Cost Calibration:** Because investigator overhead and average transaction-loss values shift due to macroeconomic conditions, the `COST_FN_RATIO` parameter is audited and updated on a regular quarterly schedule. This dynamically recalibrates decision boundaries without necessitating a full model retraining loop.
- **Retraining Label-Delay Buffer:** Transaction disputes and fraud chargebacks carry a standard 30-to-90-day dispute window. Retraining continuous pipelines on fresh data risks exposing the model to biased cohorts of apparently normal transactions where fraud has simply not been reported yet. SentinelFlow mitigates this by enforcing a **90-day temporal lookback buffer**; retrain sets only ingest matured cohorts, discarding data younger than 90 days from updates.